

1. RESEARCH QUESTION AND CORPUS

This analysis investigates sentiment and emotion patterns in Reddit comments related to gun politics. The main research question is: *What emotions dominate gun politics discussions?*

The corpus was constructed from the simplified *GoEmotions* training dataset. First, duplicate comments were removed using the comment id. Then, the corpus was filtered to include only comments from three target subreddits:

- gunpolitics,
- progun,
- liberalgunowners.

The final corpus contains 352 comments. The distribution across subreddits is uneven:

Subreddit	Number of comments
gunpolitics	68
progun	128
liberalgunowners	156

This imbalance is important for interpretation. Raw emotion counts are naturally higher in subreddits with more comments. Normalized rates are more suitable when comparing emotional density across subreddits.

2. VADER SENTIMENT RESULTS

VADER was used to calculate sentiment scores for each post. The post-level output contains four VADER columns:

- vader_positive,
- vader_negative,
- vader_neutral,
- vader_compound.

The most important value is the compound score, which summarizes the overall sentiment of a comment on a scale from strongly negative to strongly positive.

The generated VADER summary statistics should be reported from the summary CSV.

Statistic	vader_positive	vader_negative	vader_neutral	vader_compound
count	352.000	352.000	352.000	352.000
mean	0.144	0.132	0.725	-0.022
std	0.182	0.162	0.204	0.469
min	0.000	0.000	0.133	-0.902
max	0.867	0.820	1.000	0.899

The histogram of VADER compound scores should be interpreted as the overall sentiment distribution of the corpus. If most values are concentrated around zero, then the corpus is mostly neutral or mixed. If the histogram has a stronger left tail, this indicates more negative comments; if it has a stronger right tail, this indicates more positive comments.

Given the political and conflict-oriented topic of gun discussion, a mixed distribution is expected. Some comments express criticism, fear or distrust, while others contain support or agreement. VADER should not be interpreted as giving one “positive” or “negative” label for the whole corpus. The corpus is better understood as sentimentally mixed, with the exact tendency depending on the compound-score and histogram shape.

3. NRC EMOTION RESULTS

The NRC lexicon was used to identify emotion-associated words in the comments. The analysis used the following eight NRC emotion categories:

Emotion	Total count
anger	167
anticipation	101
disgust	93
fear	173
joy	85
sadness	120
surprise	56
trust	133

If the NRC sentiment categories are also considered, the corpus contains:

Category	Total count
negative	273
positive	213

The most frequent non-valence emotion is fear with 173 matches, closely followed by anger with 167 matches. This is substantively meaningful for a gun-politics corpus. Discussions about gun policy often involve security, violence, and social conflict. These topics are likely to activate words associated with fear and anger.

The relatively high count of trust is also important. It means that words associated with the NRC trust category appeared frequently. It suggests that the corpus is not only hostile or negative.

The lowest count is surprise. This suggests that the corpus is less dominated by unexpectedness or shock than by threat-related and conflict-related language.

Overall, the NRC results suggest that the corpus is emotionally mixed but leans toward negative and threat-related language, especially fear and anger.

4. SUBREDDIT COMPARISON

The NRC emotion counts by subreddit are:

Emotion	gunpolitics	progun	liberalgunowners
anger	31	68	68
anticipation	15	49	37
disgust	21	31	41
fear	28	77	69
joy	18	38	29
sadness	23	51	46
surprise	9	29	18
trust	18	68	47

The most frequent NRC emotion differs slightly across the subreddits:

Subreddit	Most frequent NRC emotion
gunpolitics	anger
progun	fear
liberalgunowners	fear, closely followed by anger

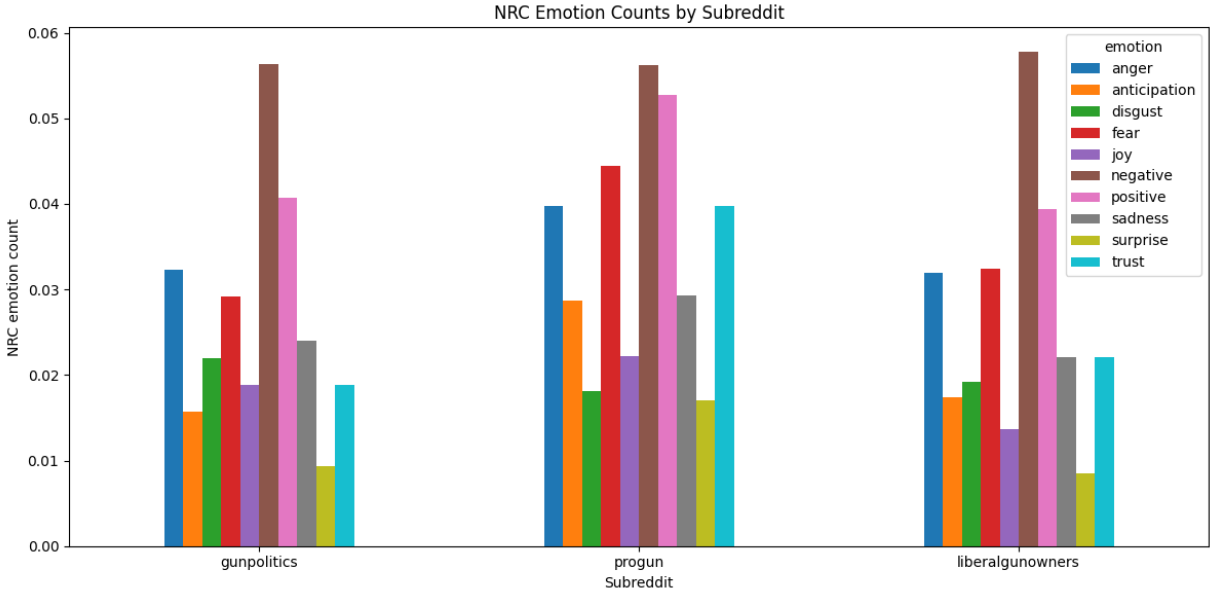
In gunpolitics, anger is the most frequent emotion category. This fits the subreddit’s explicitly political framing: discussions may focus on legislation, government action, perceived unfairness, rights violations, or ideological conflict.

In progun, fear is the most frequent category, followed by anger and trust. This pattern suggests that the subreddit contains not only oppositional or angry language, but also language related to security, protection, and group confidence. The high trust count may reflect pro-gun arguments connected to responsibility, rights, lawful ownership, or community identity.

In liberalgunowners, fear is also the most frequent category, with anger almost equal. This is interesting because this subreddit combines gun ownership with a different political identity than the more traditionally conservative gun spaces. The high fear and anger counts may reflect discussion of threats from multiple directions: gun violence, political extremism, state policy, public stereotypes, or intra-political disagreement.

However, because the subreddits contain different numbers of comments, raw counts alone should not be overinterpreted. liberalgunowners has 156 comments, while gunpolitics has only 68. Therefore, larger counts in liberalgunowners partly reflect the larger amount of text. For a fairer comparison, the normalized NRC rates per token are more appropriate.

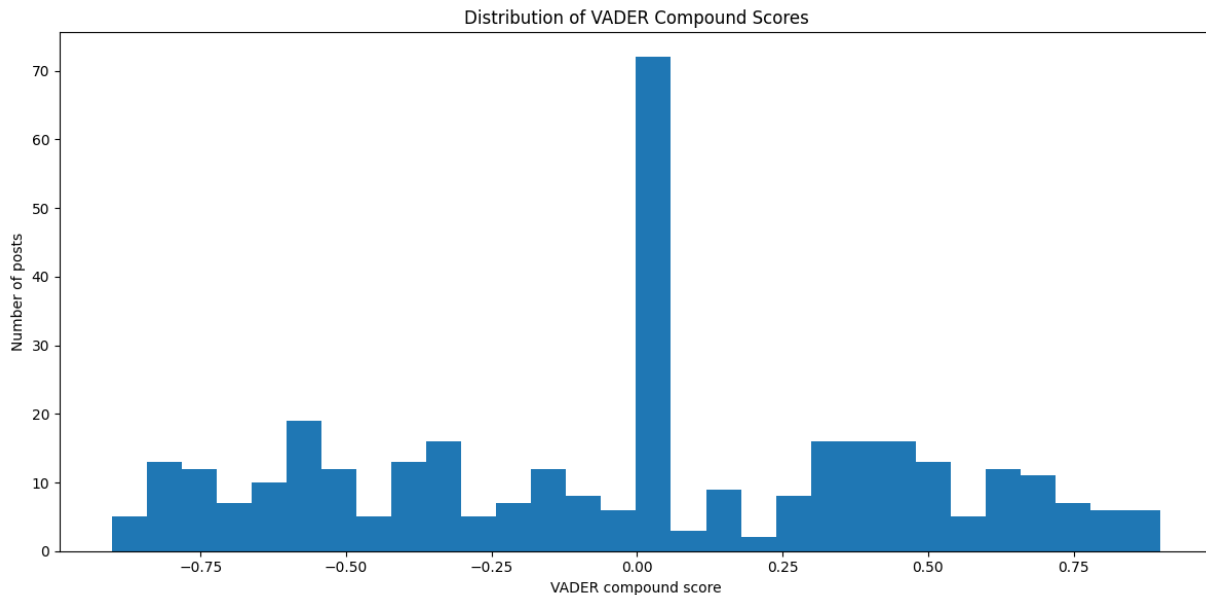
5. VISUALIZATION INTERPRETATION



The bar chart of total NRC emotion counts shows that fear and anger are the most prominent non-valence emotions in the full corpus. This supports the interpretation that gun-politics discussions are strongly shaped by threat perception and political conflict. Trust and sadness also appear with notable frequency, suggesting that the emotional profile is not only aggressive but also includes concern, disappointment, responsibility, or appeals to reliable institutions and communities.

The histogram of VADER compound scores visualizes the overall sentiment distribution. This plot is useful because the mean compound score alone can hide variation. A corpus can have a near-zero mean while still containing many strongly positive and strongly negative comments. Therefore, the histogram should be used to determine whether the corpus is genuinely neutral or instead polarized/mixed.

The grouped bar chart of NRC emotion counts by subreddit shows that progun and liberalgunowners generally have higher raw emotion counts than gunpolitics. However, this is partly because they contain more comments. The chart is still useful for seeing emotion composition: progun has especially high fear and trust, while liberalgunowners shows high fear and anger. gunpolitics, despite having fewer comments, still shows a visible anger-oriented profile.



6. LIMITATIONS OF DICTIONARY METHODS

The main limitation of dictionary-based sentiment and emotion analysis is that it is based on word matching. The NRC method counts whether words appear in a predefined emotion lexicon, while VADER uses lexical and rule-based sentiment scoring. These methods are interpretable, but they do not fully understand the context.

For example, dictionary methods can struggle with negation. They can also miss sarcasm and political language creates additional problems (words such as "freedom", "control", "rights", "safety" in community-specific meaning).

Also gun-related communities may use abbreviations, memes, legal references, or ironic expressions that are not included in the NRC lexicon or are not interpreted correctly by VADER.

Another limitation concerns corpus construction. This corpus was built by selecting subreddits. If the corpus was instead constructed using other keywords, the results could change substantially. Subreddit-based sampling captures community-specific discourse, while keyword-based sampling captures topic-specific discourse across communities.

7. CONCLUSION

The gun-politics corpus contains 352 Reddit comments from three selected subreddits. The NRC results show that the most frequent non-valence emotions are fear and anger, while the valence categories show more negative than positive matches. This suggests that the corpus is strongly shaped by conflict, threat perception, and political concern.

Across subreddits, gunpolitics is most associated with anger, while progun and liberalgunowners show fear as the most frequent emotion. However, raw subreddit counts must be interpreted carefully because the number of comments differs across communities. Normalized rates are necessary for stronger comparative claims.

In general, dictionary methods provide a useful first overview of sentiment and emotion patterns, but should be interpreted with caution because they cannot fully capture sarcasm, negation, political framing, or community-specific meanings.